

* Layer Normalization *

* It is a technique to stabilize & speed up training by normalizing activations across features within a single data sample.

Helps with :-

- * Exploding / vanishing Gradients
- * Unstable training...
- * Slow convergence.

Before LN (Varies a lot) $[120, 300, 50, 900]$ $\rightarrow$ After LN (balanced) $[-0.5, 0.1, -1.2, 1.6]$

* Formula *

$$\text{LayerNorm}(x) = \gamma \cdot \frac{x - \mu}{\sqrt{\sigma^2 + \epsilon}} + \beta$$

μ = mean

σ^2 = variance

ϵ = small value ($1e-5$)

γ = learnable scale (gain)

β = learnable shift (bias)

Why important in LLMs?

- * Stable training in very deep nets.
- * Faster Convergence.
- * Better gradient flow.
- * Reduces internal covariate shift.
- * Works well with variable lengths & small batch sizes

@codingdid

